

# **DAYANANDA SAGAR COLLEGE OF ENGINEERING**

**Shavige Malleshwara Hills, Kumaraswamy Layout, Bangalore-560078**

**Department of Artificial Intelligence and Machine Learning**



## **INTERNSHIP REPORT**

**ON**

### **“AI Assistant system”**

*Submitted in partial fulfillment for the award of degree(20AI8ICINT)*

## **BACHELOR OF ENGINEERING IN**

**Department of Artificial Intelligence and Machine Learning**

*Submitted by:*

**M Guru Akhil - 1DS20AI031**

Conducted at



**Filoffee Software Pvt Ltd**

**Department of Artificial Intelligence and Machine Learning  
Dayananda Sagar College of Engineering Bangalore-78**

# DAYANANDA SAGAR COLLEGE OF ENGINEERING

Shavige Malleshwara Hills, Kumaraswamy Layout, Bangalore-560078  
Department of Artificial Intelligence and Machine Learning



## CERTIFICATE

This is to certify that the Internship titled “**AI Assistant System**” carried out by **Mr M GURU AKHIL[1DS20AI031]**, a bonafide student of Dayananda Sagar College of Engineering, in partial fulfillment for the award of **Bachelor of Engineering**, in **Artificial Intelligence and Machine Learning** during the year 2023-2024. It is certified that all corrections/suggestions indicated have been incorporated in the report.

The internship report has been approved as it satisfies the academic requirements in respect course Internship (20AI8ICINT)

**Signature of Reporting Head**  
**Gourav Terwardker**  
Team Lead  
Filoffee Software Pvt Ltd

**Signature of Internship Coordinator**  
**Prof. Kavya D N**  
Dept. of AI & ML  
DSCE

**Signature of HoD**  
**Dr. Vindhya P Malagi**  
Dept. of AI & ML  
DSCE

### External Viva:

Name of the Examiner

Signature with Date

1) \_\_\_\_\_

\_\_\_\_\_

2) \_\_\_\_\_

\_\_\_\_\_

# DAYANANDA SAGAR COLLEGE OF ENGINEERING

Shavige Malleshwara Hills, Kumaraswamy Layout, Bangalore-560078  
Department of Artificial Intelligence and Machine Learning



## DECLARATION

I, **M Guru Akhil**, final year student of Artificial Intelligence and Machine Learning, Dayananda Sagar College Of Engineering, Bangalore - 560 078, declare that the Internship has been successfully completed, in **Filoffee Software Pvt Ltd**. This report is submitted in partial fulfillment of the requirements for the award of Bachelor Degree in Artificial Intelligence and Machine Learning, during the academic year 2023- 2024.

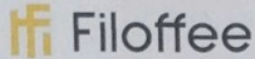
Date:

USN: 1DS20AI031

Place: Bangalore

NAME: M Guru Akhil

# OFFER LETTER



Date: 16 November 2023

## INTERNSHIP LETTER

Dear Akhil,

Filoffee is pleased to offer you the position of Intern – AI (Artificial Intelligence) in our organization based in Bangalore, Karnataka.

This letter comes effective **08 January 2024** and extends for a period of **6 months**.  
Your fixed Stipend for the internship will be **INR 20,000/- per month**.  
You will be reporting to **Mr. Sushanth Mascarenhas, Head – of Research and Development**.

We believe you will be an excellent addition to our team. We trust that your knowledge, skills, and experience will be among our most valuable assets, and we look forward to having you onboard. You will be required to work from our corporate office based out of Bangalore.

Post your internship, you can apply for employment at Filoffee, on the basis of your performance. Filoffee shall absorb you as an Employee.

As such your internship will include induction, on-the-job training, and project exposure and focus primarily on learning and developing new skills and gaining a deeper understanding of concepts through the hands-on application of the knowledge you learned in class.

The project details and technology platform will be shared with you on or before the commencement of training.

Once again, congratulations and we look forward to working with you.

Yours Sincerely,  
For Filoffee

Shilpa B H Gowda  
Manager – Human Resources

Member Name M. Guru Akhil  
Date 16/11/2023

Signature M. Guru Akhil

**Filoffee Software Private Limited**

#L4 & L29, 2<sup>nd</sup> Floor, Pine Platinum, 2<sup>nd</sup> 'A' main road, HSR Layout, Sector 6, Bangalore, Karnataka- 560102

# COMPLETION CERTIFICATE

## A C K N O W L E D G E M E N T

This Internship is a result of accumulated guidance, direction and support of several important persons. We take this opportunity to express our gratitude to all who have helped us to complete the Internship.

We express our sincere thanks to our Principal **Dr. B G Prasad**, for providing us adequate facilities to undertake this Internship.

We would like to thank **Dr. Vindhya P Malagi**, Head of Department – AI & ML, for providing us an opportunity to carry out Internship and for her valuable guidance and support.

We would like to thank **Mr Gourav Terwardker**, Filoffee Software Pvt Ltd, for guiding us during the period of internship.

We express our deep and profound gratitude to our Internship Co-ordinator, **Prof. Kavya D N**, Assistant Professor – AI & ML, for her keen interest and encouragement at every step in completing the Internship.

We would like to thank all the faculty members of AI & ML department for the support extended during the course of Internship.

We would like to thank the non-teaching members of the Department of AI & ML, for helping us during the Internship.

Last but not the least, we would like to thank our parents and friends without whose constant help, the completion of Internship would have not been possible.

**M GURU AKHIL [1DS20AI031]**

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# CHAPTER 1

## INTRODUCTION

AI Assistants represent a groundbreaking advancement in digital interaction, revolutionizing how we engage with technology. Unlike traditional software applications, AI Assistants harness the power of artificial intelligence to understand, interpret, and respond to human queries and commands in a natural language format. These assistants are designed to emulate human-like intelligence, continuously learning and improving their capabilities over time. By leveraging sophisticated algorithms and machine learning models, AI Assistants can perform a wide range of tasks, from providing personalized recommendations and assistance to automating routine tasks and facilitating complex decision-making processes.

### **1. Natural Language Interaction.**

One of the key features of AI Assistants is their ability to understand and respond to natural language queries. Users can communicate with these assistants in everyday language, eliminating the need for specific commands or technical jargon. This intuitive interface enhances user experience and accessibility, enabling seamless interaction across diverse demographics and skill levels.

### **2. Personalised Assistance:**

AI Assistants leverage data analytics and user insights to deliver personalized assistance tailored to individual preferences and needs. By analyzing user behavior, preferences, and past interactions, these assistants can anticipate user intent and provide relevant recommendations, suggestions, and solutions in real time.

### **3. Automation and Efficiency:**

AI Assistants streamline tasks and workflows by automating routine processes and simplifying complex tasks. From scheduling appointments and managing emails to conducting research and generating reports, these assistants empower users to focus on high-value activities while delegating repetitive tasks to AI-driven automation.

### **4. Continuous Learning and Improvement.**

AI Assistants employ machine learning algorithms to continuously learn from user interactions and feedback, improving their accuracy and performance over time. Through iterative training and refinement, these assistants evolve to better understand user preferences, context, and



nuances, enhancing the quality and relevance of their responses.

### **5. Integration and Connectivity:**

AI Assistants seamlessly integrate with a variety of digital platforms, applications, and devices, enabling cross-channel connectivity and interoperability. Whether accessing information from the web, interacting with third-party services, or controlling smart home devices, these assistants serve as central hubs for accessing and managing digital resources.

## CHAPTER 2

### DESCRIPTION OF ORGANIZATION

#### FILOFFEE Software Pvt Ltd



#### Introduction

Filoffee Software Private Limited (FSPL) is an Indian company that uses AI to simplify multi-vendor network maintenance and bring AI to computer networking. FSPL was incorporated in March 2022 and has its registered office in Bangalore, Karnataka. Akshay Balaganur and Sushanth Mascarenhas are directors of the company. Filoffee's AIOps platform provides network assurance and self-healing capabilities.

Its ML algorithms can detect and predict performance issues before they affect users. They believe in “**Knowledge sharing enables equal opportunity for all**”. They offer internships in various different fields such as Networking , AI and DevOps. They provide Mentor ship to help students and enable them to move forward.. They also have an alternative channel open always to help us, when we are in need or when are struck in any kind of unfavorable situations

#### Founder



Akshay is a Network Engineer by profession. His every day superhero routine involves saving

lives of LAN networks. After his 5 year stint at Cisco Systems, he now runs an independent Network consultancy firm by name Airowire Network and specializes in Backbone Network Designs. He is the Director and founder of Filoffee.AI.

He co-founded wfihero, a new age hotspot solution for hospitality sector. He brings in more than 12 years of versatile experience to build Filoffee into an innovative social enterprise.

Know more about them on: <https://filoffee.ai/>

## **Co-founder**

### **Mr. Sushanth Mascarenhas**

Sushanth is the Co-Founder & Head of Research and Development at Filoffe.AI bringing experience from previous roles at Airowire Networks and Aruba, a Hewlett Packard Enterprise company. Sushanth Mascarenhas holds a 2007 - 2011 Bachelor's Degree in Electrical, Electronics and Communications Engineering @ T John Institute Of Technology. With a robust skill set that includes Troubleshooting, Network Engineering, Networking, Network Security, Wireless Networking and more, Sushanth Mascarenhas contributes valuable insights to the industry.

### **Rakesh Ramakrishnaih**

Rakesh Ramakrishnaih, the esteemed Product Manager at Filoffee, brings a wealth of expertise and experience in the realm of Multi-Networking OEM platforms and their associated technologies. With a comprehensive understanding and hands-on proficiency in implementing Enterprise and Data Center Backbone Networks, Wireless solutions, NAC, Perimeter security measures, Traffic Management, Software-Defined WAN, and NMS solutions, Rakesh has been instrumental in optimizing network infrastructures for efficiency and performance. His tenure includes successfully leading NOC Operations for clients, where he served as the Incident, Change, and Problem Management lead, ensuring seamless operations and client satisfaction. Additionally, Rakesh has showcased his commitment to knowledge sharing and professional development by leading classroom and web-based training sessions for network consultant.

## **ABOUT THE COMPANY**

### **About Filoffee Software Private Limited**

Filoffee.AI is a product based startup registered in India and Singapore. They envision a world of self healing networks, building the future. Filoffee is a next-generation AIOps platform for network assurance.. They are on a mission to build fully autonomous networks.

### **VISION STATEMENT**

A world of self healing networks, building future.

### **MISSION STATEMENT**

Know something is wrong before your users do.

#### **Pro-active Assurance**

Their ML algorithms can detect and predict performance issues much before they impact to users..

#### **Automate Root Cause Analysis**

Their anomaly detection and correlation engine can do automated RCA suggest correlative actions.

#### **Reduce MTTR, Increase Happiness**

Network assurance doesn't have to be hard. With filoffee achieve a network with self-diagnosis and self-healing capability

### **Internship - Positions available at Filoffee for the following Profiles**

- Web Development Mobile
- Network engineering
- Content Development
- Digital Marketing
- Generative AI

## **CHAPTER 3**

### **SCOPE OF WORK**

#### **PROBLEM STATEMENT**

Developing an AI Assistant to Address WAN Link Performance Analysis Challenges:

In response to the pressing need for efficient management and optimization of Wide Area Network (WAN) links across diverse sites and groups, there arises a critical imperative to develop an AI Assistant. This AI Assistant will serve as a sophisticated solution to extract valuable insights from user data pertaining to WAN links and to analyze vendor performance based on uptime and Service Level Agreement (SLA) breaches. By harnessing the power of artificial intelligence and data analytics, this initiative aims to streamline network management processes, enhance performance monitoring capabilities, and mitigate risks associated with downtime and SLA non-compliance. The development of such an AI Assistant represents a pivotal step towards achieving operational excellence and ensuring the seamless functioning of network infrastructures in today's dynamic digital environment.

#### **Introduction to Basic Assistant System**

The need for an AI Assistant arises from the growing complexity of modern tasks and the desire for streamlined, efficient solutions. Whether in personal or professional contexts, individuals face an overwhelming volume of information and tasks, often requiring quick decisions and precise actions. An AI Assistant addresses this challenge by leveraging artificial intelligence to understand user needs, automate repetitive tasks, and provide intelligent recommendations. From managing schedules and organizing information to facilitating communication and conducting research, the AI Assistant serves as a versatile and indispensable tool, enhancing productivity and enabling users to focus on high-value activities.

During this internship, I embarked on a journey to delve into the realm of AI-driven solutions and web development, aiming to broaden my skill set and deepen my understanding of cutting-edge technologies. Inspired by my passion for innovation, I conceptualized and developed an AI Assistant empowered by the OpenAI platform, FastAPI, and React.js. This dynamic assistant

was crafted to streamline various tasks and enhance user experience across different domains, mirroring the functionalities of traditional assistants while leveraging the power of Artificial intelligence.

Through leveraging the OpenAI platform, I gained invaluable insights into the potential of natural language processing and machine learning in optimizing user interactions and automating various operations. By harnessing the capabilities of FastAPI, I learned to develop robust and scalable APIs, enabling seamless communication between frontend and backend systems. Additionally, my exploration of React.js equipped me with the tools to design responsive and intuitive user interfaces, elevating the overall user experience of the AI Assistant.

Throughout this journey, I not only mastered the intricacies of these technologies but also gained a profound understanding of web development processes, from conceptualization to deployment. By immersing myself in the development of this AI-powered assistant, I honed my skills in problem-solving, project management, and collaboration, laying a solid foundation for future endeavors in the dynamic landscape of AI-driven solutions and web development. This internship served as a catalyst for my professional growth, empowering me with the knowledge and expertise to navigate the intersection of artificial intelligence and web development with proficiency.

## **CHAPTER 4**

### **LEARNING OBJECTIVES**

#### **1. Mastery of Aruba Basics:**

The company aims to equip interns with a comprehensive understanding of Aruba networking technologies and solutions. Interns will delve into the fundamentals of Aruba networking infrastructure, including wireless access points, controllers, switches, and network management systems. Through hands-on training and guided learning modules, interns will gain proficiency in configuring, managing, and troubleshooting Aruba networks, preparing them for roles in network administration and engineering.

#### **2. Network Terminology and Product Knowledge:**

Interns will be immersed in the world of network terminology and product knowledge, gaining familiarity with key concepts, protocols, and technologies essential for modern networking environments. From TCP/IP fundamentals to VLANs, routing protocols, and network security mechanisms, interns will develop a solid foundation in networking principles. Additionally, interns will explore the diverse range of networking products and solutions offered by the company, including switches, routers, firewalls, and network management tools, deepening their understanding of industry-standard technologies and best practices.

#### **3. Implementation of AI in Networking Operations:**

With the increasing integration of artificial intelligence (AI) and machine learning (ML) in networking operations, interns will explore various techniques and methodologies for implementing AI-driven solutions in network management and optimization. Through case studies, practical exercises, and research projects, interns will examine how AI and ML technologies can enhance network performance, automate routine tasks, and improve overall efficiency in network operations. By gaining hands-on experience with AI-powered tools and

platforms, interns will be prepared to contribute to the development and deployment of AI-driven networking solutions in real-world environments.

#### **4. Research on Open Source Language Models (LLMs):**

Interns will engage in research and exploration of Generative AI Open Source Language Models (LLMs), focusing on understanding their capabilities, limitations, and applications in natural language processing (NLP) and text generation tasks. Specifically, interns will investigate Gen AI LLMs, such as GPT-3 and its variants, exploring their architecture, training methodologies, and performance characteristics. Through experimentation and analysis, interns will assess the feasibility and potential impact of integrating Gen AI LLMs into networking operations, such as automating documentation, generating network configurations, and enhancing user interactions with AI-powered assistants.



## CHAPTER 5

### CHALLENGES AND SOLUTION

#### **Challenges in Developing an AI Assistant:**

In the development of an AI assistant with a focus on generating SQL queries related to networking terms, several non-functional challenges emerge alongside the functionalities. While meeting user requirements is essential, ensuring accuracy, maintainability, reliability, and security of the system are equally critical aspects of software engineering.

a) **Conformance to specific standards:** The AI assistant must conform to industry standards and best practices in natural language processing and database management to ensure consistency and interoperability with existing systems and protocols.

b) **Performance constraints:** The system should be optimized to handle large volumes of data and complex queries efficiently, without compromising on response times or system performance.

c) **Hardware limitations:** Consideration must be given to the hardware resources available for running the AI assistant, ensuring compatibility and scalability across different hardware configurations.

d) **Maintainable:** The AI assistant's codebase should be well-structured and modular, allowing for easy maintenance, updates, and enhancements over time.

e) **Reliable:** The system must be robust and resilient, capable of handling errors, failures, and unexpected inputs gracefully, while maintaining data integrity and system availability.

f) **Testable:** Adequate testing mechanisms should be in place to verify the accuracy and reliability of the AI assistant's SQL query generation functionality, including unit tests, integration tests, and end-to-end testing scenarios.

**Solution:**

To address accuracy issues in generating SQL queries related to networking terms, the Retrieval-Augmented Generation (RAG) model offers a promising solution. RAG combines retrieval-based and generation-based approaches, allowing the AI assistant to retrieve relevant context from a knowledge source (such as a database of networking terms) and generate SQL queries based on this retrieved context. By leveraging pre-trained language models and fine-tuning them on domain-specific data, RAG can accurately capture the nuances of networking terminology and generate contextually appropriate SQL queries. Additionally, RAG's ability to learn from user feedback and adapt to different query contexts further enhances its accuracy and effectiveness in generating SQL queries related to networking terms. Integrating RAG into the AI assistant's architecture can significantly improve the accuracy and reliability of its SQL query generation functionality, thereby addressing one of the key challenges in its development.

## **HARDWARE AND SOFTWARE REQUIREMENTS**

### **Software requirements:**

- Front End: React.js
- Data Base: MySQL
- Server: Nginx
- API: Fast-API

### **Hardware Requirements:**

- MobaXterm
- Notepad++ Editor
- Processor: Intel core i5 processor
- Memory: 15.6 GB
- Hard Disk: 40 GB

## CHAPTER 6

### ACHEIVEMENTS AND CONTRIBUTION

#### System Design:

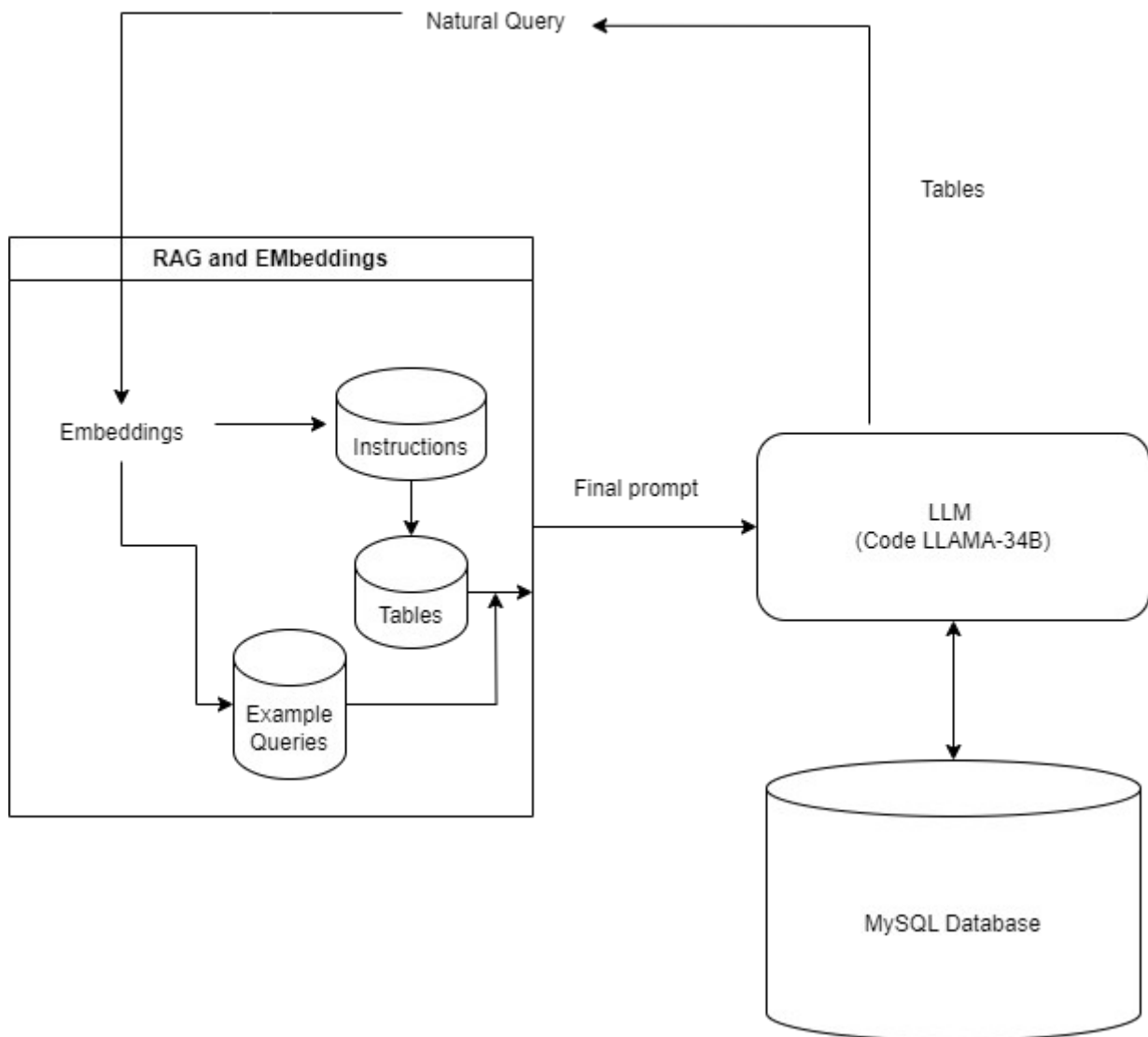


Fig. System Architecture

The AI Assistant System mainly contains:

- **RAG(Retrieval Augmented Generation)** - RAG is a state-of-the-art model developed by OpenAI that combines the capabilities of retrieval-based and generation-based approaches in natural language processing. Retrieval-based methods involve retrieving

relevant information from a large corpus of text based on user queries, while generation-based methods focus on

generating responses from scratch.

➤ **Embeddings** - Embeddings are dense vector representations of words or phrases in a continuous vector space. These representations capture semantic relationships between words and enable machines to understand and process natural language more effectively. We used embeddings mainly in prompting to retrieve required tables for prompting the text-to-sql model.

➤ **AI Assistant**- AI Assistant is a multifaceted tool designed to streamline text-to-SQL conversion and database analysis, offering users seamless access to valuable insights from complex datasets. Leveraging advanced natural language processing techniques, it interprets user queries, converting them into structured SQL queries for database retrieval. Once executed, the assistant meticulously analyzes the query results, extracting key information, identifying patterns, and generating actionable insights. By automating these processes, the assistant enhances productivity, reduces manual effort, and empowers users to make informed decisions based on data-driven intelligence

### **Flow of the Process :**

Natural Query > Embeddings > Similarity Search > Prompting > SQL query> Tables(retrieved Data)

During my internship, I spearheaded the development of an advanced AI Assistant tailored specifically for reporting tasks within the organization. This AI Assistant represented a significant milestone in enhancing reporting efficiency and accuracy, leveraging cutting-edge technologies in natural language processing and data analytics. By harnessing the power of machine learning algorithms and advanced language models, I engineered a solution capable of automating report generation, analysing complex datasets, and extracting actionable insights from large volumes of data.

## **CHAPTER 7**

### **REFLECTION AND ANALYSIS**

My internship experience provided me with invaluable insights and learning opportunities, allowing me to gain a deeper understanding of key concepts and technologies in the dynamic landscape of networking and artificial intelligence (AI). Working in a startup environment presented unique challenges and opportunities, fostering a culture of innovation, collaboration, and rapid iteration. Through hands-on experience and mentorship, I honed my skills in network basics, explored various implementations of Generative AI Language Models (LLMs), developed APIs using FastAPI, and gained proficiency in deploying applications on NGINX servers.

One of the most significant takeaways from my internship was the opportunity to delve into network basics and gain practical experience in configuring, managing, and troubleshooting network infrastructure. By working closely with experienced network engineers and administrators, I acquired a solid foundation in networking principles, protocols, and technologies, enabling me to tackle real-world challenges with confidence.

In conclusion, my internship experience was a transformative journey that allowed me to deepen my knowledge, expand my skill set, and gain practical experience in networking, AI, web development, and DevOps practices. Through reflection and analysis, I have identified areas of growth and development, paving the way for continued learning and professional advancement in the dynamic field of technology.

.

Finally, deploying applications on drone systems provided me with valuable insights into the deployment and management of video analytic applications in real-world environments. By configuring drone systems for optimal camera view, efficient data transmission, and effective PPE detection, I learned how to optimize performance, enhance safety, and ensure high availability for video analytic applications, laying the groundwork for scalable and resilient deployments in real-world scenarios.

In conclusion, my internship experience was a transformative journey that allowed me to deepen my knowledge, expand my skill set, and gain practical experience in networking, AI, web development, and DevOps practices. Through reflection and analysis, I have identified areas of growth and development, paving the way for continued learning and professional advancement in the dynamic field of technology.

## CODE

### ChatApp.js

```
import React, { useState, useEffect } from 'react';
import ChatMessage from './ChatMessage';
import './index.css';
import robot from './robot.png';
import DynamicChart from './chart'

const ChatApp = () => {
  const [messages, setMessages] = useState([ { html: 'Hello, How can I help you?', sender: 'assistant' } ]);
  const [newMessage, setNewMessage] = useState('');
  const [isTyping, setIsTyping] = useState(false);
  const [defaultSuggestions, setDefaultSuggestions] = useState([]);
  const [isUserTyping, setIsUserTyping] = useState(false);
  const [dydata,setdydata]=useState([]);
  const fetchDefaultSuggestions = async () => {
    try {
      // Make a GET request to the API endpoint for default suggestions
      const response = await fetch('http://localhost:8000/recommend');
      if (!response.ok) {
        throw new Error('Failed to fetch default suggestions');
      }

      // Parse the response as JSON
      const responseData = await response.json();

      // Extract default suggestions from the response
      const responseSuggestions = responseData.result;

      // If default suggestions exist and the user is not typing manually, set them in the state
      if (responseSuggestions && responseSuggestions.length > 0 && !isUserTyping) {
        setDefaultSuggestions(responseSuggestions);
      }
    } catch (error) {
      console.error('Error fetching default suggestions:', error.message);
    }
  };
};
```



```

useEffect(() => {
  // Fetch default suggestions when the component mounts
  fetchDefaultSuggestions();
}, [isUserTyping]);

const handleMessageSubmit = async (e) => {
  e.preventDefault();
  if (newMessage.trim() !== '') {

    const userMessage = { html: newMessage, sender: 'user' };
    const updatedMessages = [...messages, userMessage];
    setMessages(updatedMessages);
    setNewMessage('');
    setIsTyping(true);
    setIsUserTyping(false);
    // Clear default suggestions after user sends a message
    setDefaultSuggestions([]);

    try {
      // Construct the API URL with the user's message
      const apiUrl = `http://localhost:8000/query/${encodeURIComponent(newMessage)}`;

      // Make a GET request to the API endpoint
      const response = await fetch(apiUrl);
      if (!response.ok) {
        throw new Error('Failed to fetch response');
      }

      // if (response!=null)
      // {dydata=responseJson.sample_message.data;}

      // console.log('response:', response);
      if (!response.ok) {
        throw new Error('Failed to fetch response');
      }
    }
  }
}

```

```
File Edit Selection View Go Run Terminal Help
Fillofee
EXPLORER
FILOFEE
  _pycache_
  chatbot
  chat-ui
    build
    node_modules
    public
    src
      App.css
      App.js
      App.test.js
      chart.js
      ChatApp.js
      ChatMessage.js
      index.css
      index.js
      logo.svg
      reportWebVitals...
      robot.png
      setupTests.js
      .gitignore
      package-lo...
      package.json
      README.md
      chat.zip
      New WinRAR ZIP a...
      chatapp
      llama.cpp
      apikeys.txt
      assistant_1.txt
      OUTLINE
      TIMELINE
  master
  3
  8
  0
  Live Share
Ln 78, Col 49 Spaces: 2 UTF-8 CRLF {} JavaScript Go Live

chatbot > chat-ui > chat > src > JS ChatApp.js > ChatApp > handleMessageSubmit > responseHtml
8 const ChatApp = () => {
38 useEffect(() => {
41 }, [isUserTyping]);
42
43 const handleMessageSubmit = async (e) => {
44   e.preventDefault();
45   if (newMessage.trim() !== '') {
46
47     const userMessage = { html: newMessage, sender: 'user' };
48     const updatedMessages = [...messages, userMessage];
49     setMessages(updatedMessages);
50     setNewMessage('');
51     setIsTyping(true);
52     setIsUserTyping(false);
53     // clear default suggestions after user sends a message
54     setDefaultSuggestions([]);
55
56     try {
57       // Construct the API URL with the user's message
58       const apiUrl = `http://localhost:8000/query/${encodeURIComponent(newMessage)}`;
59
60       // Make a GET request to the API endpoint
61       const response = await fetch(apiUrl);
62       if (!response.ok) {
63         throw new Error('Failed to fetch response');
64       }
65
66       // if (response!=null)
67       // (dydata=responseJson.sample_message.data);
68
69       // console.log('response:', response);
70       if (response.ok) {
71         throw new Error('Failed to fetch response');
72       }
73     }
74
75     // parse the response as HTML

```

```
return (
  <div style={{ display: 'flex', flexDirection: 'column', justifyContent: 'center', alignItems: 'center', height: '80vh', overflow: 'hidden' }}>
    <div id="chat-container" style={{ height: '400px', overflowY: 'scroll', marginBottom: '10px', width: '80%', colour: 'white' }}>
      {messages.map((message, index) => (
        <ChatMessage key={index} message={message} />
      ))}
      {isUserTyping && defaultSuggestions.length > 0 && (
        <div style={{ marginBottom: '10px', display: 'flex', flexDirection: 'column', margin: '2px', width: '265px', float: 'left' }}>
          {defaultSuggestions.map((suggestion, index) => (
            <button className="btnColor" key={index} onClick={() => handleSuggestionClick(suggestion)}>
              {suggestion}
            </button>
          ))}
        </div>
      )}
    </div>
    {isTyping && <div>Typing...</div>}
  </div>

  <div style={{ display: 'flex', justifyContent: 'center', alignItems: 'center', height: '300vh', overflow: 'hidden' }}>
    <form onSubmit={handleMessageSubmit} style={{ display: 'flex', flexDirection: 'row', padding: '20px', borderRadius: '5px', width: '150%', marginTop: '-1%' }}>
      <input
        type="text"
        value={newMessage}
        onChange={(e) => {
          setNewMessage(e.target.value);
          setIsUserTyping(true);
        }}
        placeholder="Type your message..."
        style={{ width: '100%', padding: '10px', marginRight: '10px', border: '1px solid #ccc', borderRadius: '5px' }}
      />
      <button type="submit" style={{ backgroundColor: '#007bff', color: 'white', border: 'none', padding: '10px 20px', borderRadius: '5px', cursor: 'pointer' }}>Send</button>
    </form>
  </div>
  <div>
    <DynamicChart data={dydata} />
  </div>
);
```

## Chat.js

```
import React from 'react';
import Chart from 'react-apexcharts';

const DynamicChart = ({ data }) => {

  // Ensure data is not empty
  if (!data || Object.keys(data).length === 0) {
    return <div>No data provided.</div>;
  }

  // Initialize an array to store charts

  const charts = Object.entries(data).map(([key, value], index) => {
    const { name = [], xValues = [], yValues = [], chartType = 'line' } = value ?? {};

    // Ensure xValues and yValues are provided
    if (!xValues || !yValues || xValues.length !== yValues.length) {
      return null; // Skip this entry if data is incomplete
    }

    const series = [{
      name: name[0], // Using the first element of the name array as series name
      data: yValues,
    }];

    // Create options object for ApexCharts
    const options = {
      chart: {
        type: chartType || 'line',
      },
      xaxis: {
        categories: xValues,
        labels: {
          // Using the first element of the name array as x-axis label
          formatter: (value) => name[0] + ' ' + value
        }
      },
    };

    // Additional options for pie and donut charts
    if (chartType === 'pie' || chartType === 'donut') {
      delete options.xaxis;
    }

    return (
      <div key={index}>
        <Chart options={options} series={series} type={chartType || 'line'} height={350} />
      </div>
    );
  }).filter(chart => chart !== null);

  return <>{charts}</>;
}
```

## ChatMessage.js

```
import React from 'react';
// import robot from './robot.png'
const ChatMessage = ({ message }) => {
  const { html, sender } = message;
  const isUser = sender === 'user';

  const gradientColor = isUser
    ? '#43BEF7'
    : '#E7F8F9';

  return (
    (property) React.HTMLAttributes<HTMLDivElement>.style?: React.CSSProperties | undefined
    <div style={{ textAlign: isUser ? 'right' : 'left', color: sender === 'user' ? 'white' : 'black', margin: '10px' }}>
      <div
        style={{
          display: 'inline-block',
          background: gradientColor,
          padding: '10px 10px',
          borderRadius: '5px',
        }}
        dangerouslySetInnerHTML={{ __html: html }}
      />
      { /* <img src={robot} alt="Bot Icon" /> */ }
    </div>
  );
};

export default ChatMessage;
```

```

for note in notes:
    respon=get_embedding(note)
    embeddings_notes.append(respon)
embeddings_notes = np.array(embeddings_notes).astype('float32')

index_n = faiss.IndexHNSWFlat(d, M)
index_n.add(embeddings_notes)
# index_n=faiss.read_index(index,"index_notes.faiss")
distances, indices = index_n.search(query_embedding, 2)

print("Query Text:", query_text)
print("Nearest Neighbors' Indices:", indices)
print("Corresponding Distances:", distances)
matching_strings = [notes[i] for i in indices[0]]
print(matching_strings)
query_text+=instructions
query_text+="\n Important Notes: \n"
for j in matching_strings:
    query_text+=j
query_text+=tables

documents = SimpleDirectoryReader(input_files=["metadata.txt"]).load_data()

```

## **CHAPTER 8**

### **RECOMMENDATIONS**

#### **Flexible Training Schedule:**

Offer interns the flexibility to choose their working hours, accommodating their individual preferences and schedules. By allowing interns to work during times that suit them best, you empower them to maintain a healthy work-life balance and optimize their productivity.

#### **Structured Training Modules:**

Develop structured training modules and resources to guide interns through the onboarding process and provide them with a clear roadmap for skill development. These modules should cover essential topics related to the company's technologies, processes, and culture, ensuring that interns receive comprehensive training that aligns with their learning objectives.

#### **Mentorship Program:**

Implement a mentorship program that pairs each intern with an experienced mentor who can provide guidance, feedback, and support throughout their internship. Mentors play a crucial role in facilitating learning, fostering professional growth, and helping interns navigate challenges effectively. Encourage mentors to schedule regular check-ins with their mentees and provide ongoing support and encouragement.

#### **Hands-on Project Assignments:**

Assign interns to hands-on project assignments that allow them to apply their knowledge and skills in real-world scenarios. These projects should be challenging yet achievable, providing interns with opportunities to learn new concepts, solve complex problems, and demonstrate their capabilities. Encourage interns to collaborate with their peers and mentors to tackle project tasks collaboratively and share their insights and learnings.

**Feedback and Evaluation:**

Provide interns with regular feedback and evaluation sessions to assess their progress, identify areas for improvement, and recognize their achievements. Feedback sessions should be constructive, specific, and actionable, highlighting interns' strengths and areas for development. Encourage interns to reflect on their experiences, set goals for their learning journey, and take ownership of their professional development. By fostering a culture of continuous feedback and improvement, you empower interns to maximize their potential and make meaningful contributions to the company.

## **CHAPTER 9**

### **CONCLUSION**

- This application provides a powerful tool for providing insights to users, automating the reporting process, and conducting performance analysis.
- This application also helps in retrieving present details of customers regarding their down sites and links.
- This software helps in reducing the clerical work of visiting the tool repeatedly for various queries and taking the aggregation part out of context.
- This system provides fast, efficient, reliable and user-friendly interfaces in chat system and has no chance of losing data or any write operations while processing of user query i.e. customer messages.
- This software provides a good user interface such that a user of basic computer knowledge can operate the application. It also reduces effort done by the converting natural user query and also reduces the load of real time computation.



## **CHAPTER 10**

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